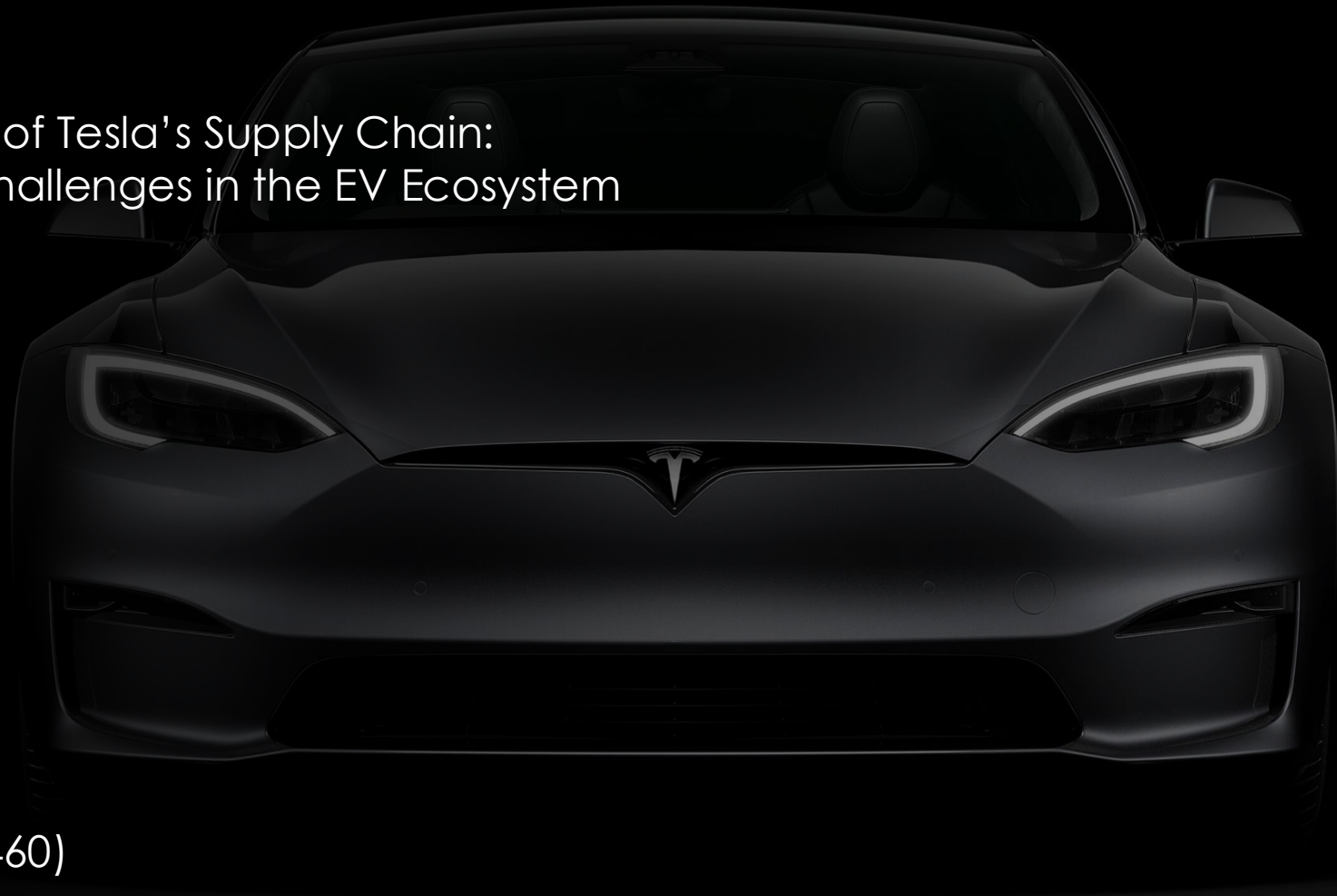


Tesla Inc



Data-Driven Analysis of Tesla's Supply Chain:
Opportunities and Challenges in the EV Ecosystem



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Matthew

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Jason

Zoe (21005998)

Executive Summary



CHALLENGES

Extreme demand
volatility & raw
material price swings

Bullwhip effect from
2020–2025 boom-bust
cycles

Geopolitical & supply
risks (chips, lithium,
cobalt, Red Sea)

KEY INSIGHTS



Low Inventory & Risk
Pooling



Smart Pricing &
Contracts

KEY QUESTION

How does Tesla use inventory management, postponement, smart pricing, and supply contracts to mitigate the bullwhip effect and build one of the most responsive automotive supply chains?

Tesla: Leading the EV Revolution



Key Performance Metrics (2024 – 2025)

\$97.7B

2024 Revenue

1.79M

2024 Vehicle Deliveries

497K

Q3 2025 Deliveries
(+7% YoY)

\$1.3T

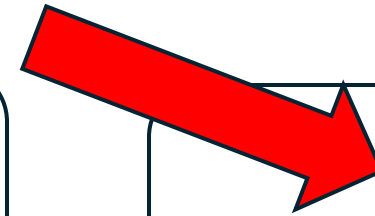
Market Capitalization
(Nov 2025)

125, 665

Employees Worldwide

6

Global Gigafactories



Note: Strong Q3 2025 performance driven by increased production capacity and strategic market expansion

Tesla's Gigafactory Network (2025)



Strategic Global Manufacturing Footprint

Fremont Factory

California, USA

Original facility, 5.3M sq ft, 10,000+ employees



Gigafactory Nevada

Nevada, USA

Batteries cells, powertrains, energy storage production



Gigafactory Shanghai

Shanghai, China

Model 3/Y for Asian Market, largest overseas facility



Gigafactory Berlin

Brandenburg, Germany

European production hub, Model Y focus



Gigafactory Texas

Austin, USA

Cybertruck & Model Y, largest facility (4.5M+ sq meters)



Gigafactory Mexico

Monterrey, Mexico

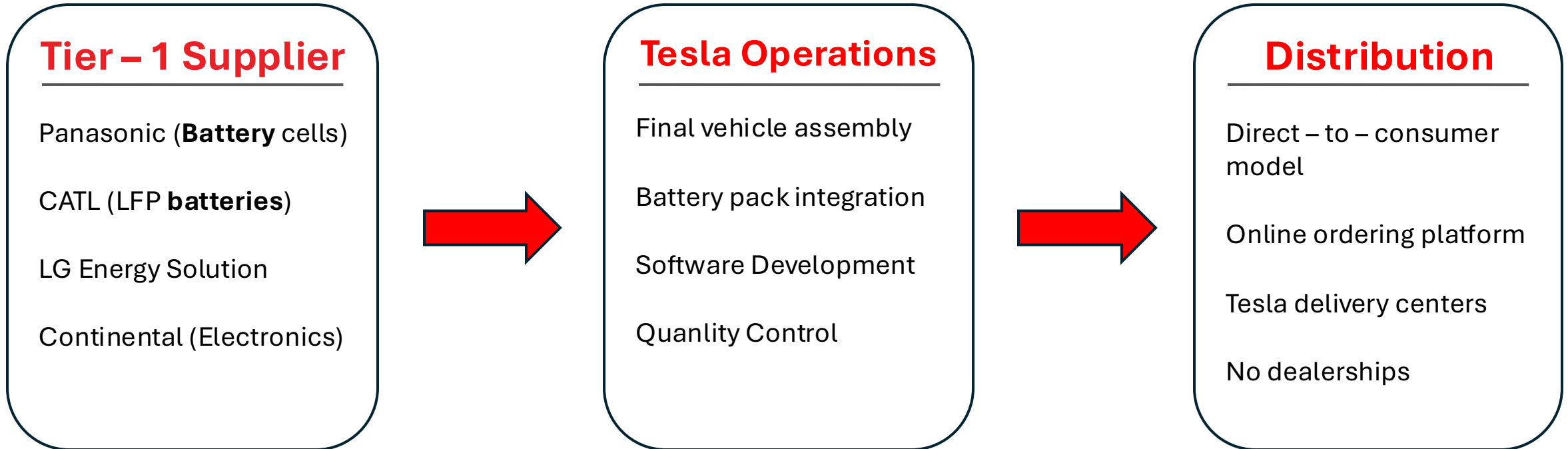
Under construction, targeting 2025 – 2026 opening



Tesla's Supply Chain Architecture



Three – Tier Integrated Structure



Key Strategy: Vertical integration with increasing in-house battery production (4680 cells)

The Battery as Core Competitive Advantage

4680 Cell Production & Cost Leadership

Cost Impact

Battery represents ~40% of vehicle price (down from 50% in 2020)

Production Milestone

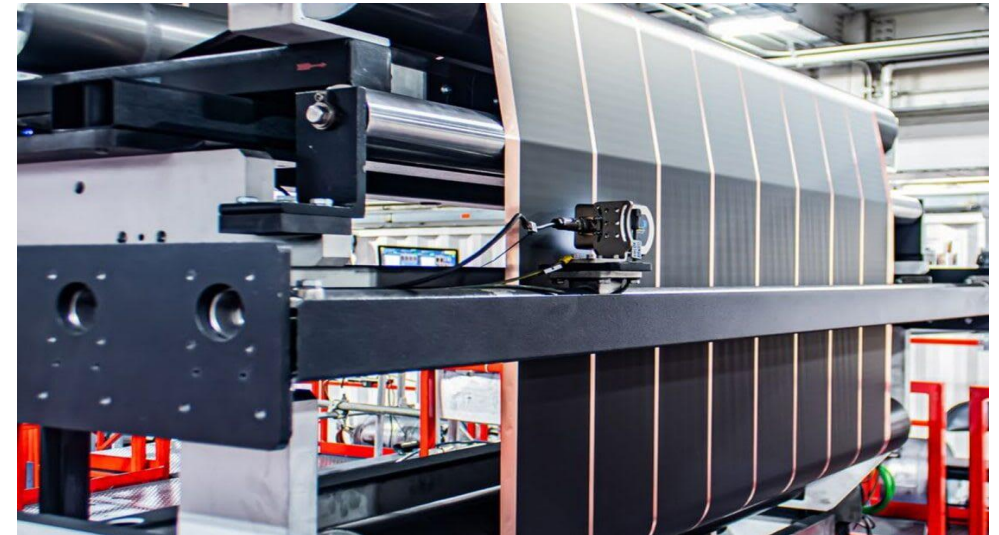
100+ million 4680 cells produced as of September 2024

Cost Advantage

4680 cells now cheapest per kWh through in-house production

Strategic Suppliers

Panasonic, CATL (LFP), LG Energy Solution partnerships



Disrupting Traditional Auto Distribution



Traditional Dealerships vs. Tesla's Direct Sales Model

Traditional Model

Dealership markup **(10 – 15%)**

Large inventory holdings **(60+ days)**

Price negotiation process

Regional price variations

Tesla's Model

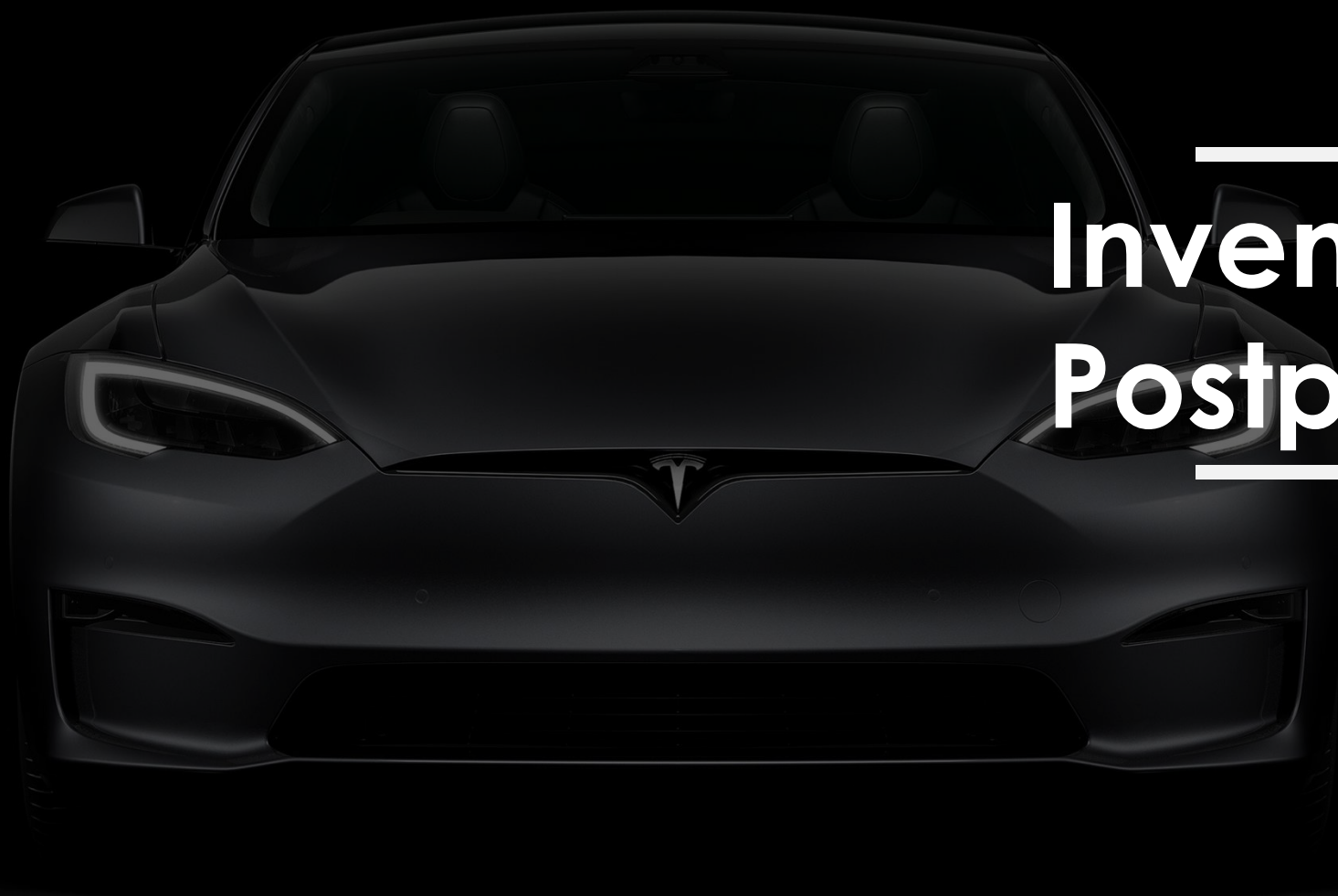
Fixed pricing nationwide

Minimal inventory **(5 – 12 days)**

Configure – to – order system

Direct customer relationships

Result: Lower costs, faster customer feedback, enhanced experience



Inventory & Postponement

Tesla's Total Inventory Days of Supply: Less Than Competitors



Year/Quarter	Tesla	Toyota	GM	Ford	Industry
2022	79	55	68	78	~70
2023	63	48	62	75	~68
2024	55	42	58	80	~72
2025 Q3	53	44	61	82	~75

Tesla's Risk Pooling in Action



1

100% Direct Online Orders



- All customer orders go straight to Tesla's central system
- No middleman (dealers) guessing or buffering demand
- Creates a single worldwide demand pool instead of hundreds of regional pools (L3 core idea)

2

3

Situational Analysis

**Inventory &
Postponement**

Bullwhip + Pricing &
Contracts

Disruptions & Outlook

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No Dealer Fragmentation



- Legacy automakers have thousands of independent dealers
- Each dealer forecasts separately → high local variability
- Tesla eliminates this fragmentation completely

3

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Demand variability aggregated across markets



- High demand in China can instantly offset low demand in Europe
- Net global variability drops dramatically
- Safety stock needed falls to only ~3–5 days (vs 30–40 days for Toyota/GM/Ford)

Situational Analysis

Inventory &
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Form Postponement in Automotive Supply Chains: Competitors vs Tesla



Competitors vs Tesla Form Postponement Strategies

Competitors: Partial/Virtual Form Postponement

- Build standardized base vehicle first
- Delay paint, wheels, interior to dealer/region
- Some accessories & OTA software at/after dealership
- Dealers cause **Bullwhip Effect** → distorted forecasts
- Still 60–80 days finished-goods inventory

Tesla: Full Configure to Order Form Postponement

- Generic “vanilla” car leaves factory
- Paint, wheels, interior, FSD chosen after production
- Most features delivered OTA post-delivery
- Only ~18 days finished inventory

Result: Tesla eliminates dealers → removes Bullwhip Effect → true form postponement → dramatically lower inventory & risk

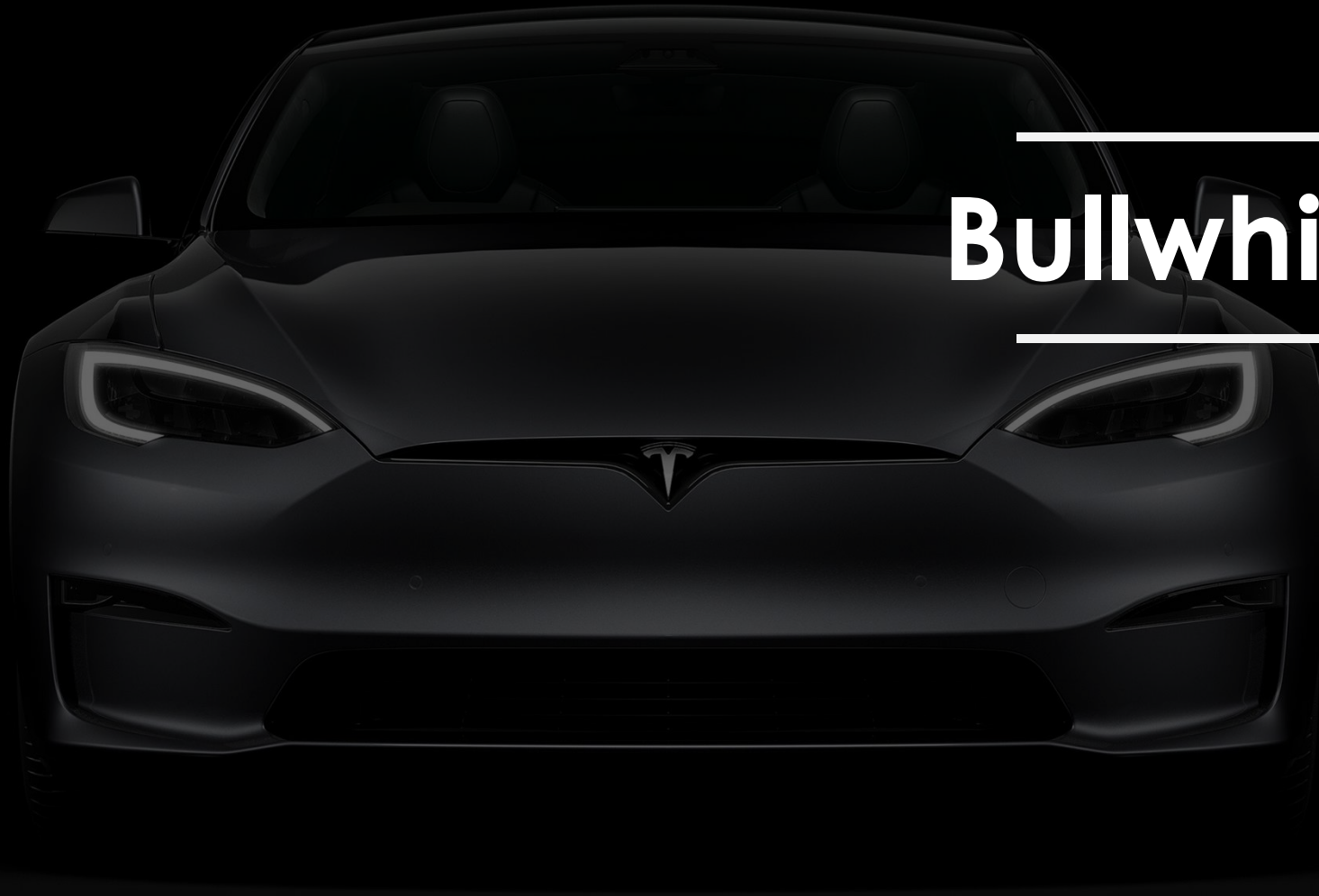
Quantitative Impact of Tesla's Strategy



Year/Quarter	Legacy OEM's	Tesla (2024-Q3 2025)	Tesla Advantage
Finished Vehicle-Inventory	60-80 days	~18 days	4–6× less stock
Total Inventory Turnover	5-7 times/year	6.7 times/year	Higher capital efficiency
Capital Tied Inventory	\$30–50 billion	~\$13 billion	~\$20–25B freed up
Obsolescence Risk	High (dealer lots)	Near zero	No model-year leftovers



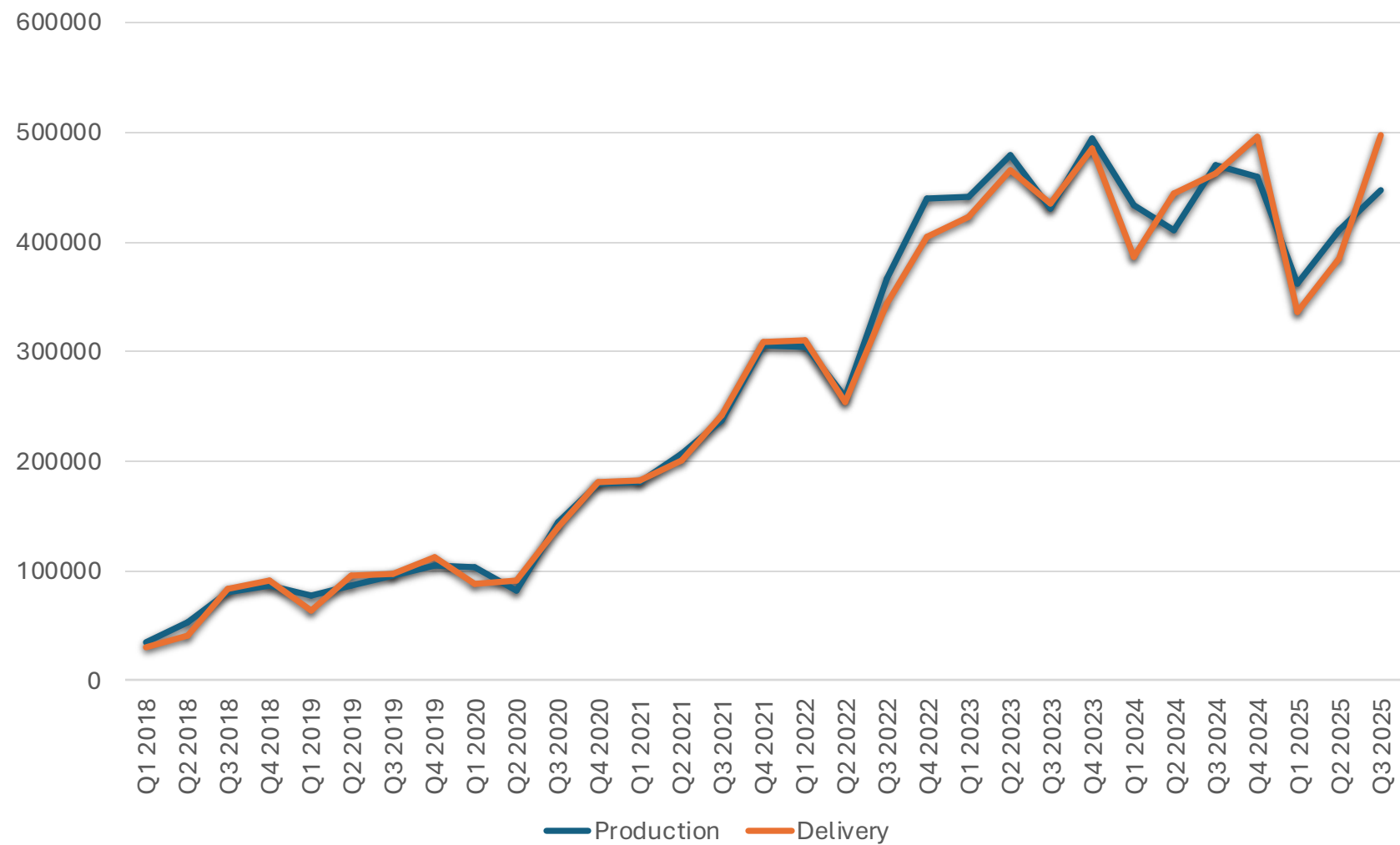
Bullwhip Effect



Tesla's Demand Volatility: 2018-2025



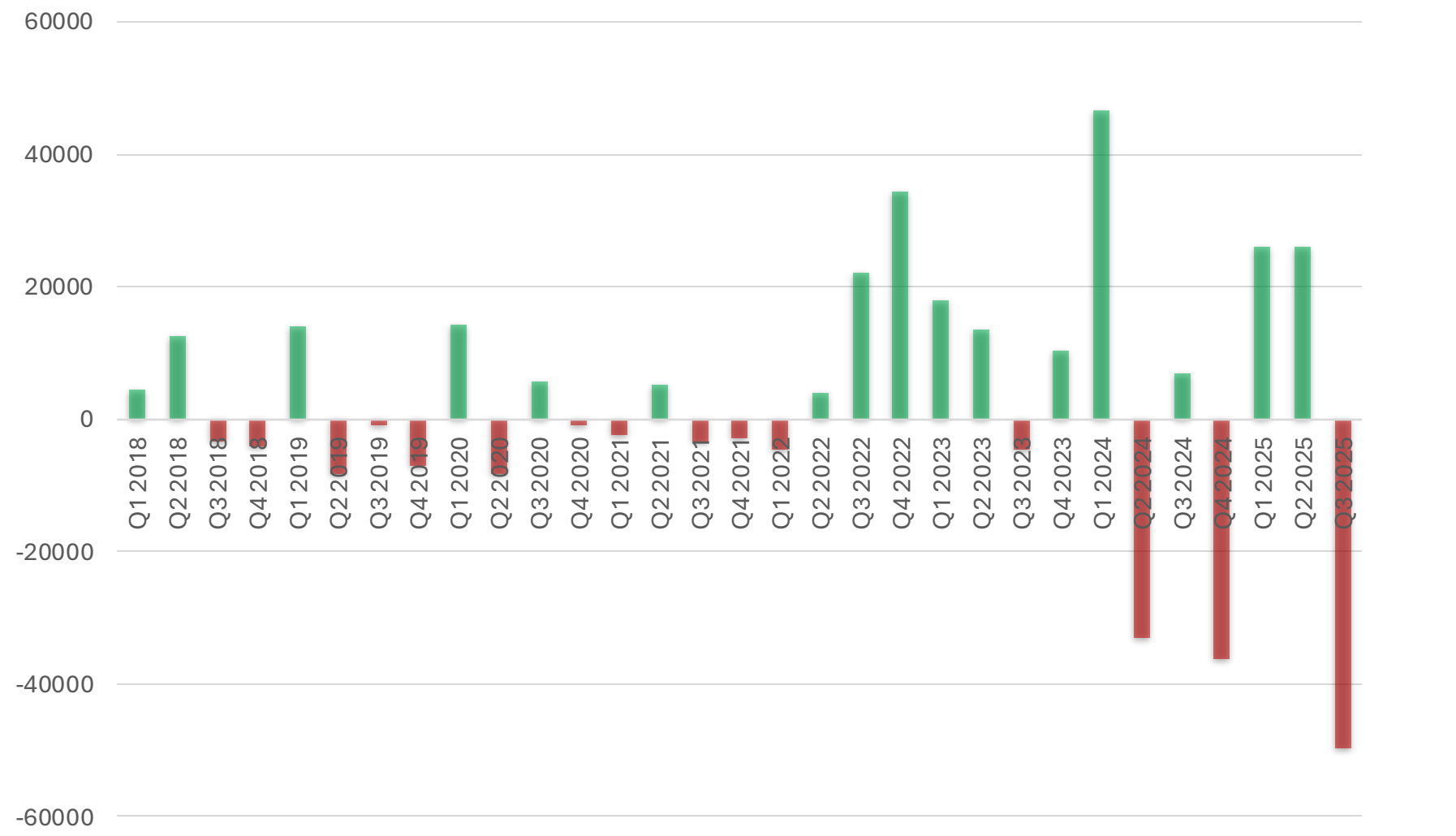
Production and Delivery



Tesla's Oscillating Bullwhip Pattern: 2018-2025



Production and Delivery gap



Tesla's Bullwhip Mitigation Strategies



1

Direct Sales Model



No dealer forecasting distortion



perfect downstream visibility

2

Real-Time Order Visibility



- track exact customer orders

3

Frequent Price Adjustments



- Acts as demand smoothing
- creates forward-buying behavior

X

External Factors Still Cause Bullwhip

-Policy changes -competition

Situational Analysis

Inventory &
Postponement

**Bullwhip + Pricing &
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Disruptions & Outlook

Despite Mitigation Efforts, Bullwhip Amplitude is Increasing

Production and Delivery gap



Situational Analysis

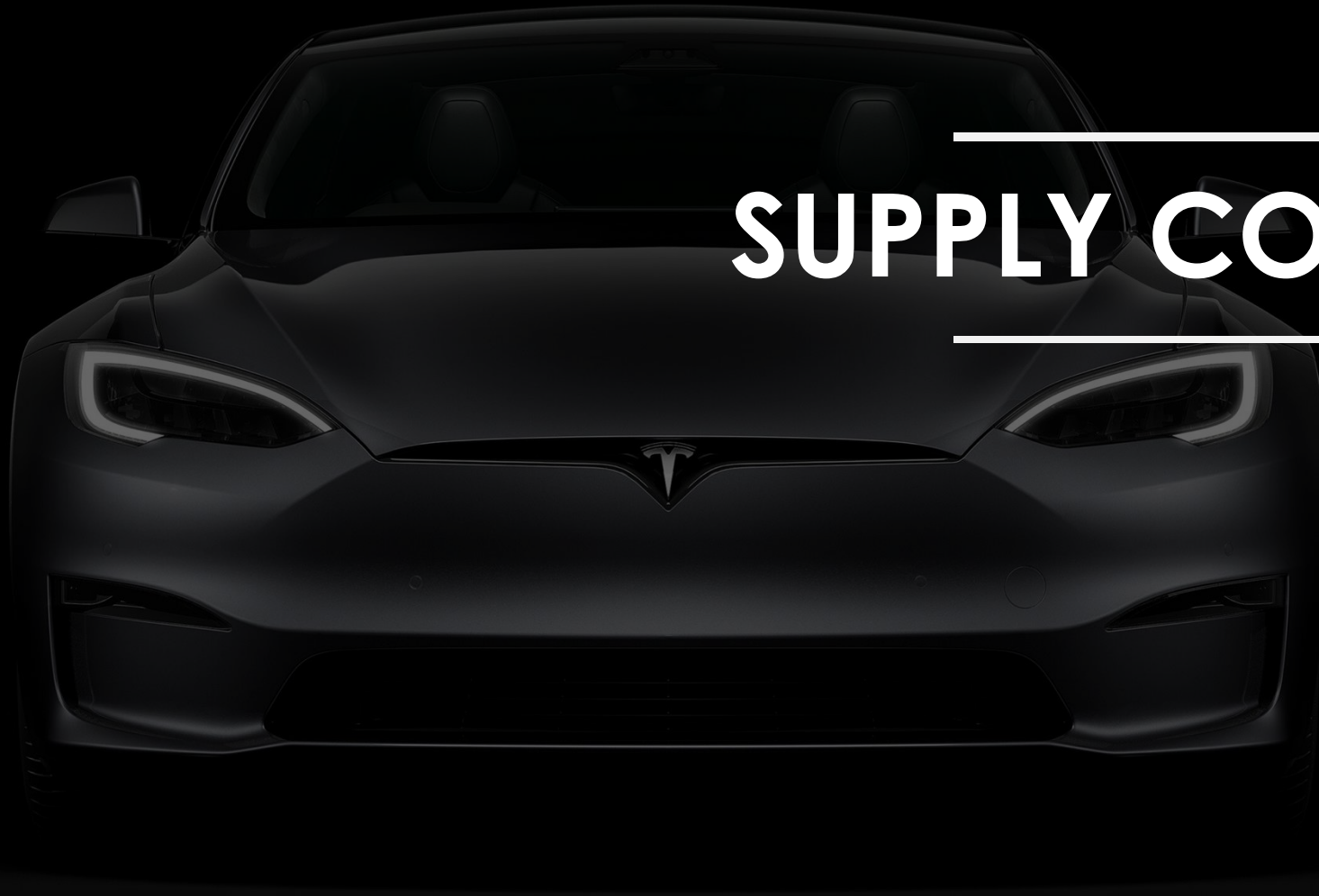
Inventory &
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**Bullwhip + Pricing &
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Disruptions & Outlook



SUPPLY CONTRAST



2021-2022 (Chip/Battery Crisis)



Dominant contract



Take-or-pay + public
"price pressure" letters



Risk bearing



Examples



Outcome



Situational Analysis

Inventory &
Postponement

**Bullwhip + Pricing &
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Disruptions & Outlook

2021-2022 (Chip/Battery Crisis)



Dominant contract



Risk bearing



Suppliers 100% (Glencore cobalt, Panasonic)



Examples



Outcome



Situational Analysis

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Disruptions & Outlook

2021-2022 (Chip/Battery Crisis)



Dominant contract



Risk bearing



Examples



Outcome



- Cancelled orders after double-booking
- Forced 10-20% price cuts



2021-2022 (Chip/Battery Crisis)



Dominant contract



Risk bearing



Examples



Outcome



Sub-optimal
(double ordering,
supplier anger)



Situational Analysis

Inventory &
Postponement

**Bullwhip + Pricing &
Contracts**

Disruptions & Outlook

2021-2022 (Post Crisis)



Dominant contract



Revenue-sharing + Cost-sharing + Quality flexibility



Risk bearing



Examples



Outcome



Situational Analysis

Inventory &
Postponement

**Bullwhip + Pricing &
Contracts**

Disruptions & Outlook

2021-2022 (Post Crisis)



Dominant contract



Risk bearing



Shared (CATL LFP
fixed-price, LG 4080
ramp-up funding)



Examples



Outcome



Situational Analysis

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Disruptions & Outlook

2021-2022 (Post Crisis)



Dominant contract



Risk bearing



Examples



Outcome



- CATL multi-year LFP deal
- Panasonic cost-plus 4680
- Rio Tinto 5-year lithium off-take



Situational Analysis

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2021-2022 (Post Crisis)



Dominant contract



Risk bearing



Examples



Outcome



Supply-chain
coordinated →
both sides win



Situational Analysis

Inventory &
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Disruptions & Outlook

Observation

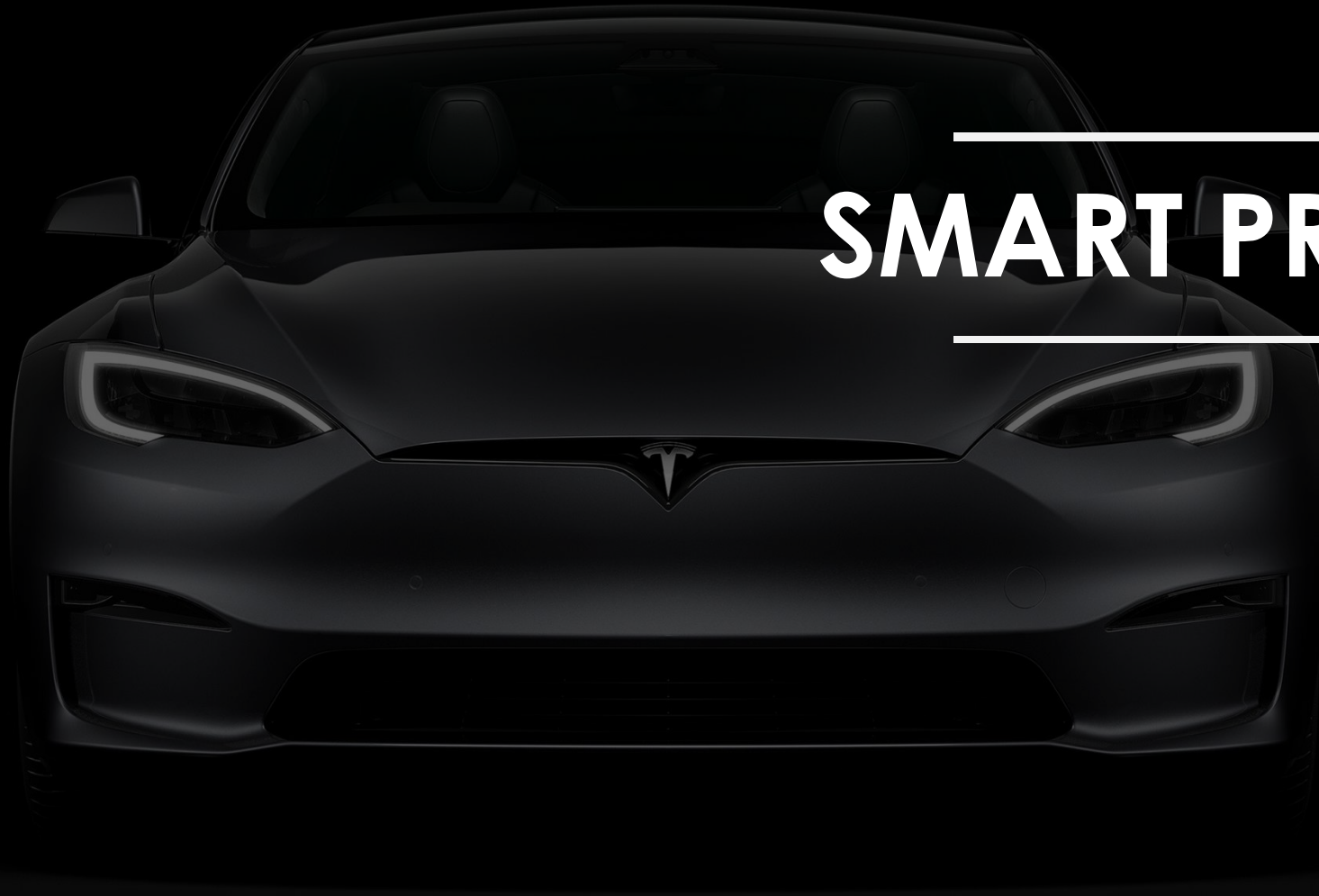


- When supply is tight → transfer risk to supplier
- When supply stabilizes → switch to buy/ revenue-sharing/ cost-sharing contracts → total profit ↑ and suppliers willingly invest billions





SMART PRICING



Demand Curve Shifting



Action	Repeated Price Cuts	End-of-quarter flash sales	Hold price when factories full
Concept	Markdown pricing	Time-based differentiation	Stop markdowns when utilization high
Result	Volume +50% YoY	Inventory days < 10	Gross margin recovered to 20%+



Situational Analysis

Inventory &
Postponement

**Bullwhip + Pricing &
Contracts**

Disruptions & Outlook

Observation



- Use dynamic pricing not to “lose money”, but to fill gigantic fixed capacity (Shanghai, Berlin, Texas) and maximize total supply-chain profit

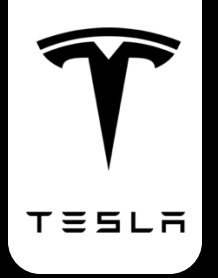


Situational Analysis

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Disruptions & Outlook



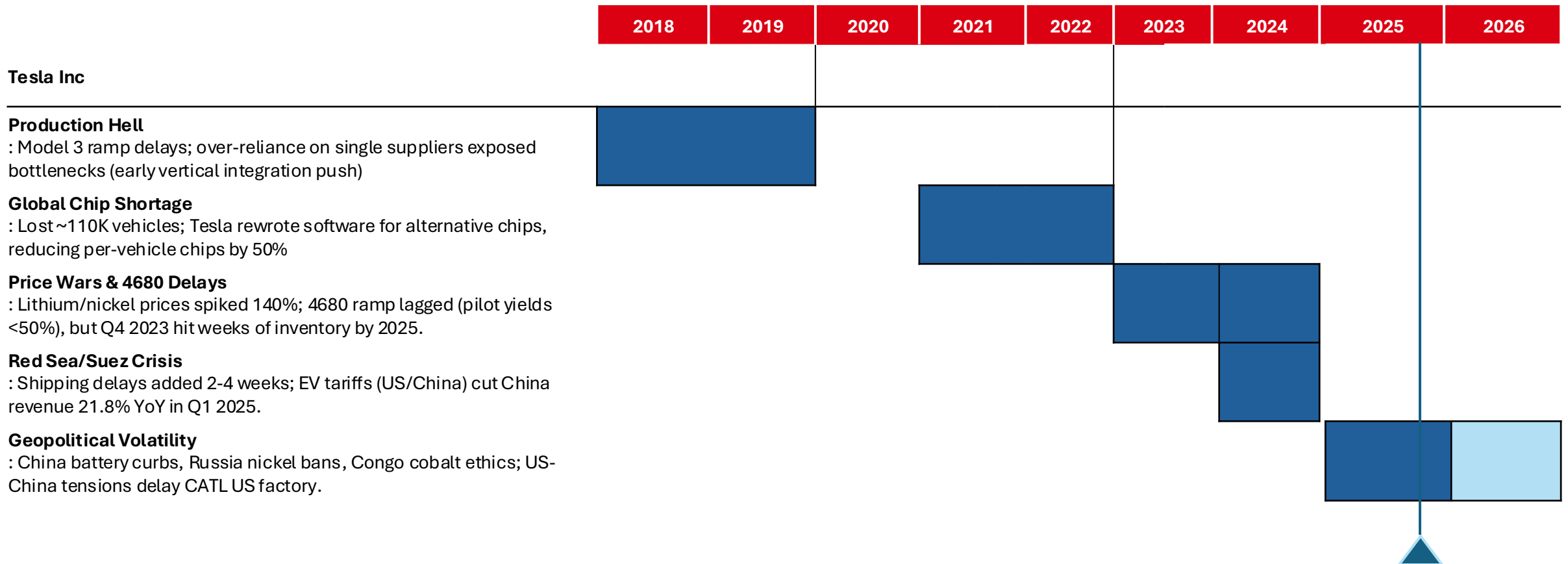
Disruptions and Outlook



Key Supply Chain Disruptions: Tesla's Battleground



Major Disruptions Timeline (2018–2025)



Situational Analysis

Inventory &
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Disruptions & Outlook

How Tesla Responded & Evolved Its Supply Chain Design



1

Crisis Rewrites & Agile Fixes



- Chip shortage 2021–22 → Rewrote firmware in days to use alternative chips
- 4680 delays → Fixed yields and now ahead of Cybertruck demand
- Red Sea 2024 → Instantly rerouted ships + added buffer stock

2

3

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Vertical Integration & Hedging



- 80% in-house (batteries, software, castings)
- Direct mining deals + multi-supplier mix (CATL, LG, Panasonic, Rio Tinto)
- Battery recycling recovers 30%+ lithium

3

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Situational Analysis

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Inventory &
Postponement

3

Global Risk Pooling & Responsiveness



- 6+ regional Gigafactories → Geographic risk pooling
- Mexico + India factories coming 2026–27 → Reduce China exposure
- Direct-to-consumer + real-time orders → True demand visibility (no dealer distortion)

Disruptions & Outlook

Bullwhip + Pricing &
Contracts

Recommendations & Key Takeaway



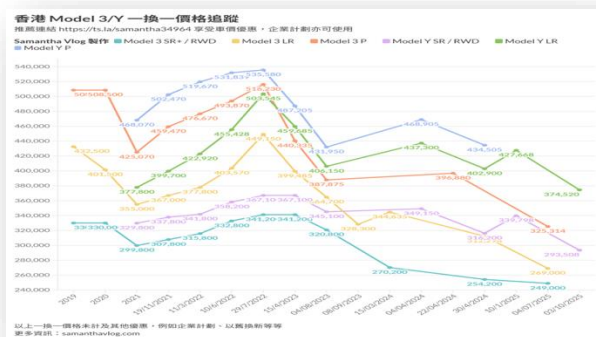
Accelerate Geographic Risk Pooling



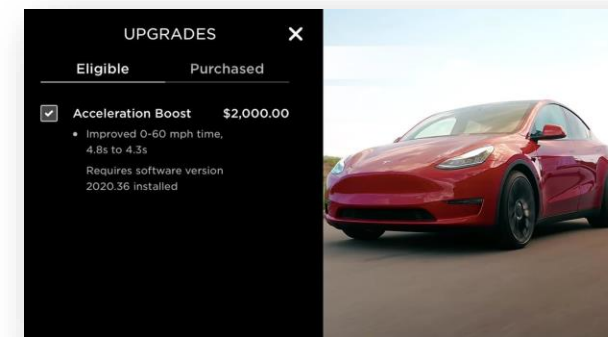
Shift to Option Contracts



Refine Dynamic Pricing Algorithm



Push Postponement Further



Situational Analysis

Inventory & Postponement

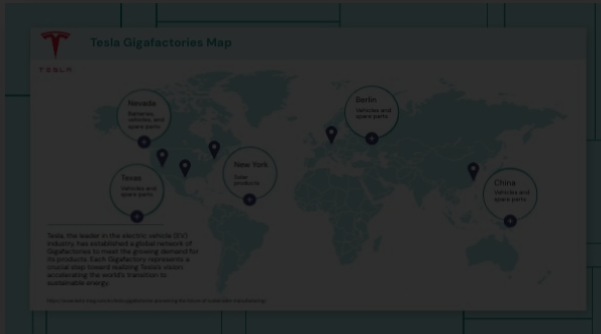
Bullwhip + Pricing & Contracts

Disruptions & Outlook

Recommendations & Key Takeaway



Accelerate Geographic Risk Pooling

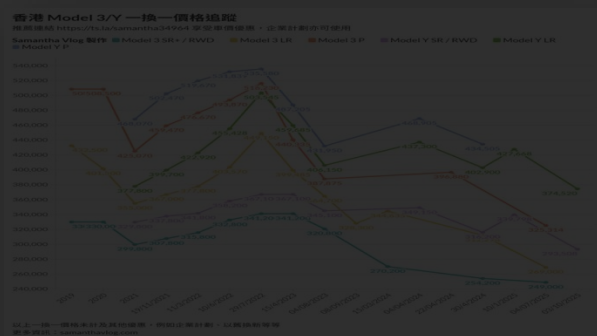


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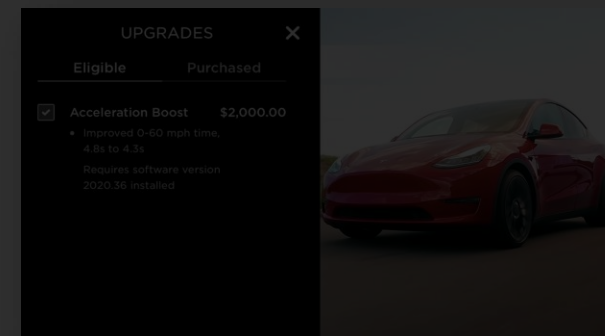


Q&A

Refine Dynamic Pricing Algorithm



Push Postponement Further



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